LESSON 2.8

Describe Data Using the Mean

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can determine the mean of a data set, use a target mean to find a missing value, and explain the mean as the fair share and the balance point of the data.

WHY THIS MATTERS Every week, Miguel walks dogs after school to earn money, and he keeps his earnings in a notebook. Across 4 days he earned $60 in all — but not the same amount each day. Later this lesson you'll help Sofia, who tracks her points every basketball game and wants to finish the season averaging 15 points per game.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Mean *Spanish: Media*	The average of a data set. Add all the values, then divide by how many values there are.	Add, then divide by the count
Average *Spanish: Promedio*	Another word for the mean — the amount each value would be if the total were shared equally.	A typical amount
Measure of center *Spanish: Medida de centro*	A single number used to describe the middle, or typical value, of a whole data set.	Mean and median are both
Balance point *Spanish: Punto de equilibrio*	The value where the total distance of the data above it equals the total distance below it. The mean is the balance point.	The mean balances the set
Data set *Spanish: Conjunto de datos*	The whole collection of values that were gathered or measured.	8, 15, 11, 6
Target mean *Spanish: Media objetivo*	The mean a person is trying to reach.	A goal average of 29 points.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Four friends have 3, 5, 6 and 10 stickers. Find the mean.

1. Add: 3 + 5 + 6 + 10 = 24.
2. Count: there are 4 values.
3. Divide: 24 ÷ 4 = 6. Check: 6 is between 3 and 10.

Answer: 6 stickers each

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

New data set: a dog walker earns $9, $11, $16, and $12 over 4 days. First, what is the total?

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The total of all the values is _____ .
- There are _____ values, so I divided by _____ .
- The mean is _____ , which means _____ .
- The mean must fall between _____ and _____ .

Word bank mean | average | total | sum | divide | count | fair share | balance point | center | typical | data set | value

WATCH OUT!

A common mistake

The most common mistake with the mean is adding correctly and then dividing by the wrong number

LESSON 2.8

Describe Data Using the Mean

Level 0 · Most support · every hint printed

MASTERY CHECK

- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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WHY THIS MATTERS Every week, Miguel walks dogs after school to earn money, and he keeps his earnings in a notebook. Across 4 days he earned $60 in all — but not the same amount each day. Later this lesson you'll help Sofia, who tracks her points every basketball game and wants to finish the season averaging 15 points per game.

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1 **Circle the letter of the best answer. Show how you know.**

Find the mean of: 4, 6, 8, 10

(A) 6

(B) 7

(C) 8

(D) 28

STEPS TO TRY

1. The mean is what each value would be if the total were shared out equally among all four numbers.
2. Mean = total ÷ how many values. Count first: there are 4 values here.
3. Step 1: add 4 + 6 + 8 + 10 to get the total. Step 2: divide that total by 4.

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

2 **Circle the letter of the best answer. Show how you know.**

Miguel earned $60 walking dogs over 4 days. What is his mean daily earning?

(A) $4
(B) $15
(C) $20
(D) $60

STEPS TO TRY

1. A typical day is the amount Miguel would earn if he earned the SAME amount all 4 days.
2. You already know the total, so only one step is left: split that total equally among the days.
3. Divide the $60 total by the 4 days to find the equal share for one day.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Find the mean of: 3, 5, 7, 9, 11

(A) 5
(B) 7
(C) 9
(D) 35

STEPS TO TRY

1. Mean means average: level all five numbers out so they are all the same.
2. Find the total of all five numbers first, then divide by the number of values, 5.
3. Step 1: add 3 + 5 + 7 + 9 + 11. Step 2: divide that total by 5.

Put the data in order _____

WORK

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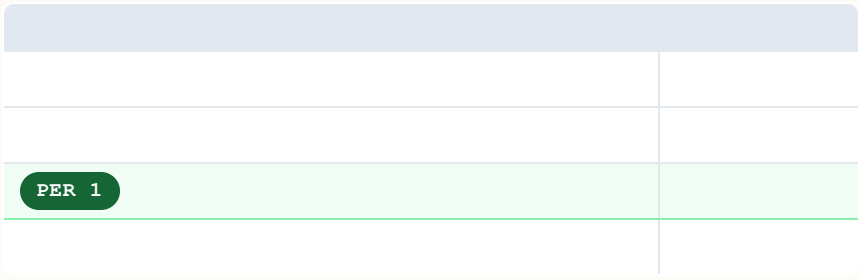
4 **Circle the letter of the best answer. Show how you know.**

In her first 4 games, Sofia scored 12, 20, 16, and 11 points. What is her mean points per game so far?

(A) 14.75
(B) 15
(C) 20
(D) 59

STEPS TO TRY

1. The mean here answers: how many points per game has Sofia averaged so far?
2. Add up the four game scores, then divide the total by the number of games played, 4.
3. Step 1: add 12 + 20 + 16 + 11. Step 2: divide that total by 4 — the answer will not be a whole number.



1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

EXPLAIN YOUR THINKING

Four games are below 15 and four games are above 15. Does that already mean the data balances at 15? What else would you have to measure?

SENTENCE STARTER Counting games is not enough because ___. To balance, I have to measure how ___ each score is from ___, and then compare the total ___ to the total ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help